Math 432/532: Introduction to Topology II

HW #7

1. Let $f : S^n \to S^n$ be a map with $\deg(f) \neq 1$. Show that there is a point $x \in S^n$ with $f(x) = -x$.

2. Let $f : S^n \to S^n$ be a map with n even. Show that there is a point $x \in S^n$ with $f(f(x)) = x$.

3. Show that, if n is odd, $-\mathrm{id}_{S_n} \sim \mathrm{id}_{S_n}$. (Hint: Try doing it when $n = 1$ first.)

4. Let K be a nonempty connected simplicial complex, and suppose that $f : |K| \to |K|$ is homotopic to a constant map. Show that there exists $x \in |K|$ with $f(x) = x$.

5. Suppose that $f : |K| \to |K|$ is simplicial and that the set $S := \{x \mid f(x) = x\}$ is finite. You may assume that every element of S is a vertex of K^1, which we stated but did not prove earlier in the course. More precisely, $S = \{\hat{A} \mid A \in K, f(A) = A\}$.

   (i) Show that, if $f(A) = A$, then there is no proper face $B \subsetneq A$ such that $f(B) = B$.

   (ii) Suppose that $f(A) = A$. Show that one can order the vertices of A $(v_0, \ldots, v_i)$ such that $f(v_j) = v_{j+1}$ for all $0 \leq j \leq i - 1$ and $f(v_i) = v_0$.

   (iii) Let $\sigma = (v_0, \ldots, v_i)$ be the orientation of A that you constructed in part (ii). Show that $f_i(\sigma) = (-1)^i \sigma$.

   (iv) Show that $\Lambda_f = |S|$.